SL No. 3 OE 1 /Sem 6/I REVISION 2021

SIXTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING

AND TECHNOLOGY

SOLAR POWER TECHNOLOGIES

MODEL QUESTION PAPER SET 1

Time: 3 Hour *Max.Marks: 75*

PART A

I Answer **all** questions in one word or one sentence. Each question carries 1 mark.
(9*1 = 9 marks)

1	Name a solar energy to electrical energy conversion system.	M1.01	R
2	Name any one application where solar radiation is converted to heat energy.	M1.02	R
3	Explain the function of a solar collector.	M2.01	U
4	Name the type of solar collector used in applications which require a temperature range of 40 to 100 degree Celsius.	M2.02	R
5	Name one factor to be considered for deciding the tilt angle of a PV module.	M3.01	R
6	Explain the manner in which solar panels are connected to form an array.	M3.03	U
7	Explain Solar PV array.	M3.03	U
8	Name the basic building unit of a solar PV module.	M4.01	R
9	Explain how a bypass diode is connected with respect to a solar cell.	M4.01	U

PART B

II. Answer any **eight** questions from the following, each question carries 3 marks.

(8*3 = 24 marks)

1	Explain the potential of solar energy to address environmental issues related to the use of Conventional energy sources.	M1.01	U
2	Draw the schematic diagram of a dish type solar cooker.	M1.03	R
3	List a few general characteristics of concentrating collectors.	M2.01	R
4	Explain the working of a Flat plate collector.	M2.02	U
5	Draw the schematic diagram of a parabolic dish power plant.	M2.04	R
6	Illustrate the working of a solar cell	M3.01	U
7	Illustrate the equivalent circuit of a solar cell and indicate various current components.	M3.02	U
8	List the standard ratings of a typical solar PV module.	M3.04	R
9	List the types of batteries that can be used in a stand-alone Solar Photovoltaic System.	M4.02	R
10	Show the block diagram of a standalone Solar PV system and identify the blocks.	M4.04	U

PART C

Answer **ALL** questions. Each question carries 7 marks.

(6*7 = 42 marks)

III	Compare active and passive solar heating.	M1.04	U
	OR		
IV	Explain the working of a solar water heater using natural circulation system.	M 1.03	U
V	Illustrate and explain Solar drying.	M 1.03	U
	OR		
VI	Illustrate the schematic arrangement and explain the operation of a Solar Air heater.	M 1.03	U
VII	Show the working of a Line focusing parabolic collector with the help of a neat diagram.	M 2.03	R
	OR		
VIII	Show the construction of Fresnel reflectors with the help of a neat diagram.	M 2.04	R

IX	A solar photovoltaic module having an area of 1.5 meter square gives a current and voltage of 8A and 28 V respectively, at maximum power point. The short circuit current of the module is 10 A and open circuit voltage is 32V. Find fill factor, maximum power point and efficiency of the solar cell.	M 3.02	A
	OR		
X	Identify the variations in output voltage and current of a PV module with the changes in a) Irradiance b) Cell temperature	M 3.04	A
XI	Explain the purpose of bypass and blocking diodes in solar PV systems	M 4.01	U
	OR		
XII	Explain the purpose of charge controllers and commonly used set points.	M 4.01	U
XIII	Explain the working of a Solar-wind hybrid system with help of a block diagram.	M 4.04	U
	OR		
XIV	Classify solar PV systems based on grid connectivity and storage.	M 4.04	U

BLUE PRINT

Mark Distribution

Module	Hr / Module	(hi / ∑Hi) * 123	TYPE OF QUESTIONS							
			PART A		PART B		PART C		TOTAL	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
I	14	29.69	2	2	2	6	4	28	**8**	**36**
II	13	27.57	2	2	3	9	2	14	**7**	**25**
III	15	31.81	3	3	3	9	2	14	**8**	**26**
IV	16	33.93	2	2	2	6	4	28	**8**	**36**
Total	**58**	**123**	**9**	**9**	**10**	**30**	**12**	**84**	**31**	**123**

Cognitive Level Wise Question Analysis

Mark Distribution

Cognitive Level	% Marks	Marks	TYPE OF QUESTIONS							
			PART A		PART B		PART C		TOTAL	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
R	30	36.9	5	5	5	15	2	14	**12**	**34**
U	50	61.5	4	4	5	15	8	56	**17**	**75**
A	20	24.6	0	0	0	0	2	14	**2**	**14**
Total	**100**	**123**	**9**	**9**	**10**	**30**	**12**	**84**	**31**	**123**

Question Wise Analysis

Q.No	Module Outcome	Cognitive Level	Marks	Time
I.1	M1.01	R	1	2
I.2	M1.02	R	1	2
I.3	M2.01	U	1	2
I.4	M2.02	R	1	2
I.5	M3.01	R	1	2
I.6	M3.03	U	1	2
I.7	M3.03	U	1	2
I.8	M4.01	R	1	2
I.9	M4.01	U	1	2
II.1	M1.01	U	3	7
II.2	M1.03	R	3	7
II.3	M2.01	R	3	8
II.4	M2.02	U	3	8
II.5	M2.04	R	3	7
II.6	M3.01	U	3	8
II.7	M3.02	U	3	7
II.8	M3.04	R	3	7
II.9	M4.02	R	3	7
II.10	M4.04	U	3	7
III.	M1.04	U	7	17
IV.	M 1.03	U	7	17
V	M 1.03	U	7	17
VI	M 1.03	U	7	17
VII	M 2.03	R	7	17

VIII	M 2.04	R	7	17
IX	M 3.02	A	7	17
X	M 3.04	A	7	17
XI	M 4.01	U	7	17
XII	M 4.01	U	7	17
XIII	M 4.04	U	7	17
XIV	M 4.04	U	7	17
Total			123	295

SL No. 3 OE 1 /Sem 6/II REVISION 2021

SIXTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING

AND TECHNOLOGY

SOLAR POWER TECHNOLOGIES

MODEL QUESTION PAPER SET 2

Time: 3 Hour *Max.Marks: 75*

PART A

I. Answer **all** questions in one word or one sentence. Each question carries 1 mark.

(9*1 = 9 marks)

1	Name the type of collectors used in solar to thermal energy conversion, where higher temperatures are required.	M1.02	R
2	Name the type of solar cooker which is able to generate high temperatures.	M1.03	R
3	Name the type of collector used in a low temperature solar power plant.	M2.02	R
4	Name the type of reflector suitable for high temperature solar power plants.	M2.04	R
5	State the principle behind the working of a Solar Cell	M3.01	R
6	Explain fill factor with respect to a solar cell.	M3.02	U
7	Give an example of an alkaline battery used in a Solar PV system.	M4.01	R
8	Explain THD.	M4.02	U
9	Explain Boost converter.	M4.02	U

PART B

II. Answer any **eight** questions from the following, each question carries 3 marks.

(8*3 = 24)

1	Explain the potential of Solar energy in meeting our present and future energy demands in three points.	M1.01	U
2	Compare concentrating and non-concentrating type solar collectors.	M1.01	U
3	List any three characteristics of Passive Solar heating.	M1.04	R
4	Classify concentrated solar power plants on the basis of temperature.	M2.01	U
5	Draw the schematic diagram of a Central receiver solar power plant.	M2.04	R
6	How is a solar PV module different from a solar PV cell? How can a solar PV module be made using individual solar cells?	M3.01	R

7	Draw the PV curve and VI curve of a photovoltaic module	M3.02	U
8	A Solar PV module has rated power of 100 W at STC. Its Voc = 22V and Isc = 5.75 A. Find out the fill factor. If the radiation intensity changes to 400 W/m2, will there be a change in the rated power output of the PV module? If yes, mention whether the rated power increases or decreases.	M3.03	U
9	Explain the main features of Net metering.	M4.03	U
10	Draw a neat block diagram of a solar wind hybrid system.	M4.04	R

PART C

Answer ALL questions. Each question carries 7 marks.

(6*7 = 42 marks)

III	Explain the operation of a forced circulation solar water heating system with the help of a neat schematic diagram.	M1.03	U
	OR		
IV	Explain Solar space cooling with the help of a neat schematic diagram.	M1.04	U
V	Illustrate the operation of a solar cooker.	M1.03	U
	OR		
VI	Explain the working of a solar water heater using natural circulation system.	M1.03	U
VII	Explain the working of a Solar Pond.	M2.02	U
	OR		

VIII	Explain the working of a Solar powered Stirling Engine.	M2.04	U
IX	The Fill factor of a PV panel is 0.7. Draw the PV characteristics and mark Vmp if Voc= 32V, Isc= 10A, Imp = 8A.	M3.02	A
	OR		
X	A solar photovoltaic module having an area of 1.62 meter square gives a current and voltage of 7.83A and 29.4V respectively, at maximum power point. The short circuit current of the module is 8.52A and open circuit voltage is 36.7V. Find fill factor, maximum power point and efficiency of the solar cell.	M3.02	A
XI	Explain MPPT with the help of a block diagram.	M4.02	U
	OR		
XII	Explain the working of a Buck-boost type DC to DC converter used in a solar photovoltaic power plant with the help of circuit diagrams.	M4.02	U
XIII	Explain stand-alone Solar PV systems with the help of a neat block diagram.	M4.04	U
	OR		
XIV	Explain grid connected Solar PV systems with neat block diagrams. Also mention one advantage of grid connected Solar PV systems.	M4.04	U

BLUE PRINT

Mark Distribution

Module	Hr / Module	(hi / ∑Hi) * 123	TYPE OF QUESTIONS							
			PART A		PART B		PART C		TOTAL	
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II	13	27.57	2	2	2	6	2	14	**6**	**22**
III	15	31.81	2	2	3	9	2	14	**7**	**25**
IV	16	33.93	3	3	2	6	4	28	**9**	**37**
Total	**58**	**123**	**9**	**9**	**10**	**30**	**12**	**84**	**31**	**123**

Cognitive Level Wise Question Analysis

Mark Distribution

Cognitive Level	% Marks	Marks	TYPE OF QUESTIONS							
			PART A		PART B		PART C		TOTAL	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
R	30	36.9	6	6	4	12	0	0	**10**	**18**
U	50	61.5	3	3	6	18	10	70	**19**	**91**
A	20	24.6	0	0	0	0	2	14	**2**	**14**
Total	**100**	**123**	**9**	**9**	**10**	**30**	**12**	**84**	**31**	**123**

Question Wise Analysis

Q.No	Module Outcome	Cognitive Level	Marks	Time
I.1	M1.02	R	1	2
I.2	M1.03	R	1	2
I.3	M2.02	R	1	2
I.4	M2.04	R	1	2
I.5	M 3.01	R	1	2
I.6	M3.02	U	1	2
I.7	M4.01	R	1	2
I.8	M4.02	U	1	2
I.9	M4.02	U	1	2
II.1	M1.01	U	3	7
II.2	M1.01	U	3	7
II.3	M1.04	R	3	8
II.4	M2.01	U	3	7
II.5	M2.04	R	3	8
II.6	M3.01	R	3	7
II.7	M3.02	U	3	8
II.8	M3.03	U	3	7
II.9	M4.03	U	3	7
II.10	M4.04	R	3	7
III.	M1.03	U	7	17
IV.	M1.04	U	7	17
V	M1.03	U	7	17
VI	M1.03	U	7	17

VII	M2.02	U	7	17
VIII	M2.04	U	7	17
IX	M3.02	A	7	17
X	M3.02	A	7	17
XI	M4.02	U	7	17
XII	M4.02	U	7	17
XIII	M4.04	U	7	17
XIV	M4.04	U	7	17
Total			123	295

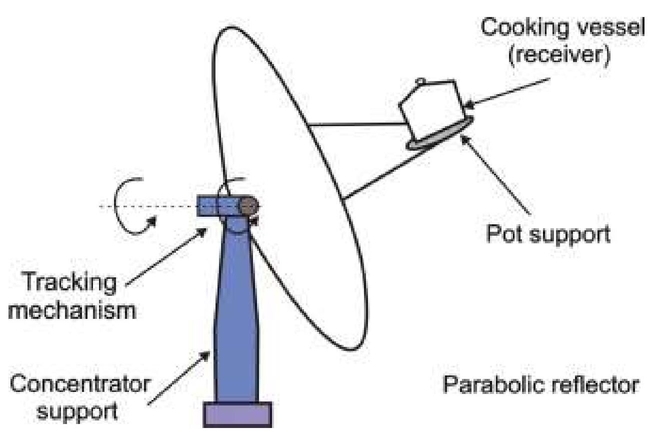
SL No. 3 OE 1 /Sem 6/I

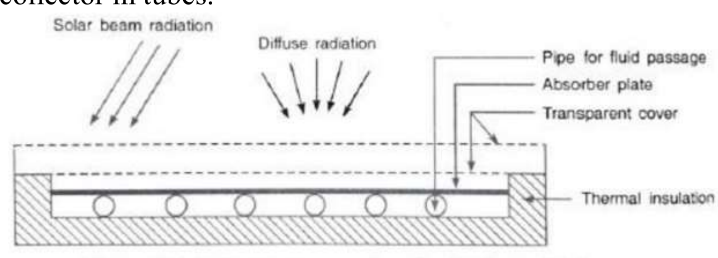
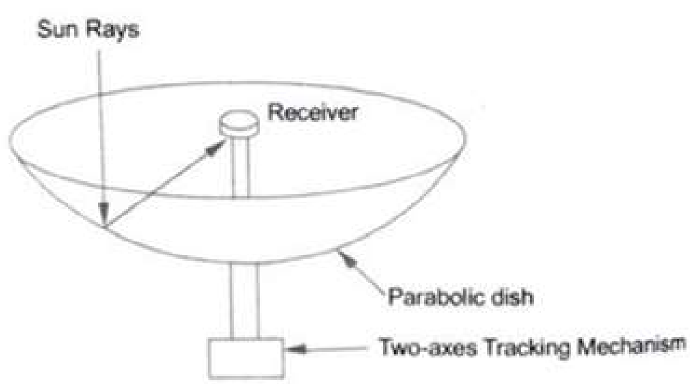
Scoring Indicators

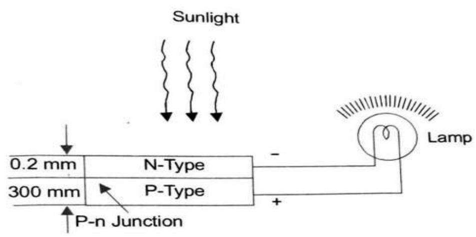
Model Question Paper I

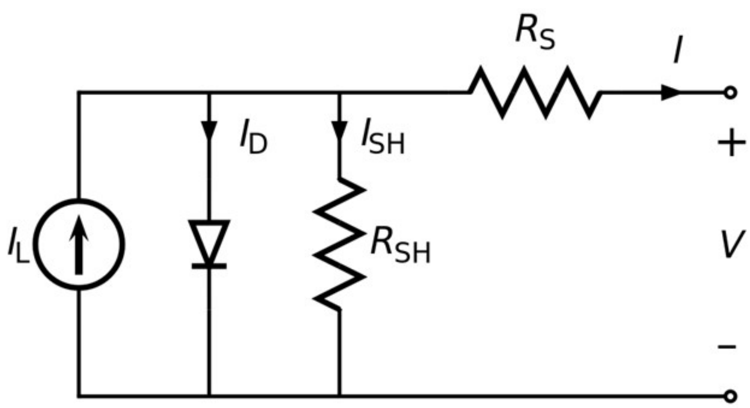
SOLAR POWER TECHNOLOGIES

Q No	Scoring Indicators	Split score	Sub Total	Total Score
	PART A			**1**
I. 1	Solar photovoltaic cell			1
I. 2	Solar water heater /Solar cooker			1
I. 3	Conversion of Solar Energy to thermal energy			1
I. 4	Flat- plate collectors			1
I. 5	Location of solar panel/Season of the year			1
I. 6	Series - parallel			1
I.7	Solar PV array is a combination of Photovoltaic modules or panels connected together in series, parallel or series-parallel.			1
I. 8	Photovoltaic cell			1
I. 9	Parallel			1
	PART B			
II. 1	Explain the potential of solar energy to address environmental issues related to the use of Conventional energy sources.			3

	Conventional sources of energy include Coal, Petroleum, Oil, Thermal Power Plants, Large Hydroelectric Power plants, etc. Use of these conventional sources of energy causes environmental pollution and degradation. Thermal power plants release toxic gases into the atmosphere. Construction of large hydroelectric power plants results in submerging of land areas and change in landscape. Besides, a lot of people will have to be relocated. Use of solar energy neither produces environmental pollution nor environmental degradation, since it neither releases toxic gases into the atmosphere nor change the original landscape of an area. Besides, solar energy is a renewable energy source. By shifting from using conventional sources of energy to solar energy environmental issues faced today can be overcome.	1 1 1		3
II. 2	Draw the schematic diagram of a dish type solar cooker.			3
		3		3
II. 3	List a few general characteristics of concentrating collectors.			3

<table>
<tr><td></td><td>1. In a concentrating collector, concentration of solar radiation is achieved using a reflecting arrangement of mirrors or a refracting arrangement of lenses.
2. The solar radiations will be concentrated on to an absorber of small area, which is usually surrounded by a transparent cover.
3. Design of a concentrating collector is more complex as compared to a flat plate collector, since a concentrated collector involves the use of a tracker to track the sun.
4. Maintenance of a concentrating collector is difficult.
(Any three of the above)</td><td>1
1
1</td><td></td><td>3</td></tr>
<tr><td>II. 4</td><td>Explain the working of a Flat plate collector.</td><td></td><td></td><td>3</td></tr>
<tr><td></td><td>Flat-plate collectors are commonly used in solar water-heating systems in residential, commercial and industrial projects. A flat-plate collector consists basically of an insulated metal box with a glazed glass cover and a dark colored absorber plate. Heat from the sun strikes the absorber plate and is transferred to a fluid that circulates through the collector in tubes.

</td><td>1.5
1.5</td><td></td><td>3</td></tr>
<tr><td>II.5</td><td>Draw the schematic diagram of a parabolic dish power plant.</td><td></td><td></td><td>3</td></tr>
<tr><td></td><td>
</td><td>3</td><td></td><td>3</td></tr>
<tr><td>II. 6</td><td>Illustrate the working of a solar cell</td><td></td><td></td><td>3</td></tr>
</table>

	Photovoltaic cell (PV cell or solar cell) is a specialized semiconductor p-n junction diode that converts visible light, Infrared (IR) or Ultraviolet (UV) radiations into direct current (DC). In semiconductors atoms carry four electrons in the outer valence shell, some of which can be dislodged to move freely in the materials if extra energy is supplied. Then, a semiconductor attains the property to conduct the current. In a solar cell the extra energy is supplied by solar radiation. This is the basic principle on which the solar cell works and generates power.  	1 1 1		3
II. 7	Illustrate the equivalent circuit of a solar cell and indicate various current components.			3

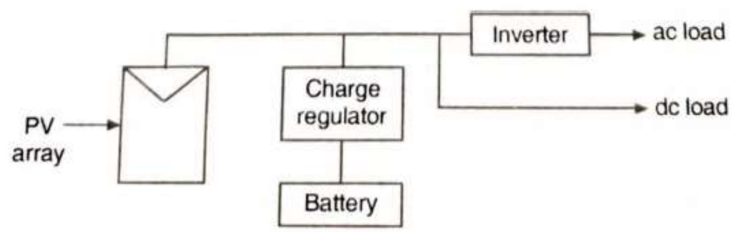
<table>
<tr><td></td><td>

I = output current (ampere)
I_L = photogenerated current (ampere)
I_D = diode current (ampere)
I_{SH} = shunt current (ampere). </td><td>2

1</td><td></td><td>3</td></tr>
<tr><td>II. 8</td><td>List the standard ratings of a typical solar PV module.</td><td></td><td></td><td>3</td></tr>
<tr><td></td><td> a. The size of an individual cell varies from 10 cm to 100 cm and a module contains about 20 to 40 cells.
b. A standard module constituting 30 cells, each of 7.5 cm diameter, can provide electrical parameters of 12 volts, 1.2 ampere, and 18 watt peak power.
c. Single PV modules of capacities ranging from 10 Wp (peak watt) to 120 Wp can provide power for different loads. </td><td>1

1

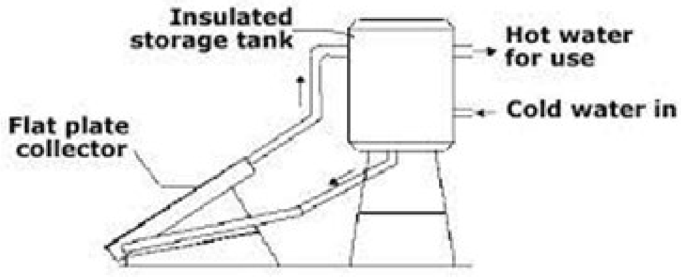
1</td><td></td><td>3</td></tr>
<tr><td>II.9</td><td>List the types of batteries that can be used in a stand-alone Solar Photovoltaic System.</td><td></td><td></td><td>3</td></tr>
</table>

	1. Lead-acid battery 2. Lithium-ion battery 3. Nickel-Cadmium(Ni-Cd) battery	1 1 1		3
II.10	Show the block diagram of a standalone Solar PV system and identify the blocks.			3
		3		3
	PART C			
III	Compare active and passive solar heating.			7

	The Major **Difference Between Active and Passive Solar Heating** is that **Active Heating** takes sunlight, either as heat or electricity, to augment heating systems. Whereas, **Passive Heating** takes heat from the sun as it comes into your home through windows, roofs, and walls to heat objects in your home.	3		7

Passive Heating	Active Heating
This system operates without pumps, blowers, or other mechanical devices.	In this system, pumps, blowers, or other mechanical devices require to circulate the working fluid for transportation of heat.
Special building design is necessary.	Special building design is not necessary.
These systems are suitable where there is ample winter sunshine and an unobstructed southern exposure is possible.	Active system can be employed at almost any location and type of building.
It is less expensive than the active system to construct and operate.	It is more expensive than the passive system to construct and operate.

	(table above)	4		
	OR			
IV	Explain the working of a solar water heater using natural circulation system			7

<table>
<tr><td></td><td>Figure shows a water heating system using natural circulation technique.

Small capacity natural circulation water heating system
The two main components of the system are the liquid flat – plate collector and the storage tank, the tank being located above the level of the collector. As water in the collector is heated by solar energy, it flows automatically to the top of the water tank and it is replaced by cold water from the bottom of the tank. Hot water for use is withdrawn from the top of the tank. Whenever this is done, cold water automatically enters the bottom. An auxiliary-heating system is sometimes provided for use on cloudy or rainy days.</td><td>4
3</td><td></td><td>7</td></tr>
<tr><td>V</td><td>Illustrate and explain Solar drying.</td><td></td><td></td><td>7</td></tr>
</table>

<table>
<tr>
<td></td>
<td>One of the traditional uses of solar energy has been for drying of agricultural products. Traditionally, drying is done on open ground. The disadvantages associated with this are that the process is slow and that insects and dust get mixed with the product. The use of dryers helps to eliminate these disadvantages. Drying can then be done faster and in a controlled manner fashion. In addition, a better quality product is obtained.
Figure given below shows a typical solar dryer.</td>
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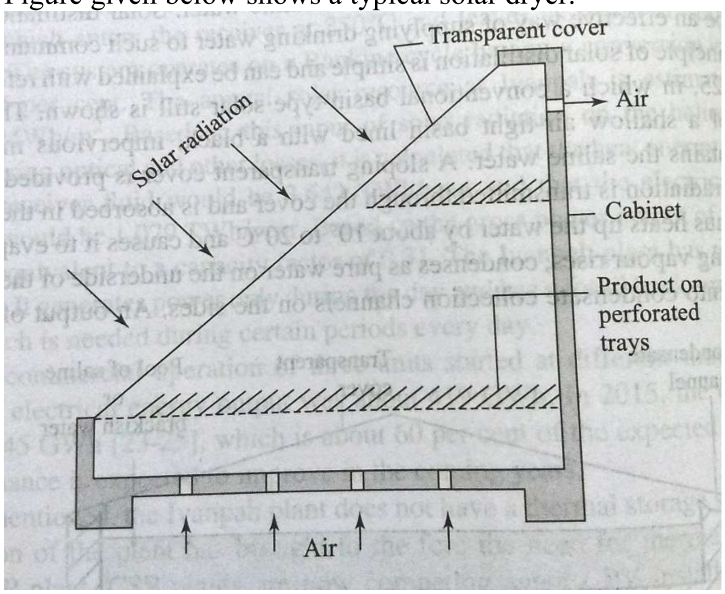
<td>
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<td>3</td>
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<td></td>
<td>It consists of an enclosure with a transparent cover. The material to be dried is placed on perforated trays. Solar radiation entering the enclosure is absorbed in the product itself and the surrounding internal surfaces of the enclosure. As a result, moisture is removed from the product and the air inside is heated. Suitable openings at the bottom and top ensure a natural circulation. Temperatures ranging from 50 to 80 degree Celsius are usually attained and the drying time ranges from 2 to 4 days. Typical products which can be dried in such devices are dates, apricots, chillies, grapes, etc.</td>
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<td>OR</td>
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<tr>
<td>VI</td>
<td>Illustrate the schematic arrangement and explain the operation of a Solar Air heater.</td>
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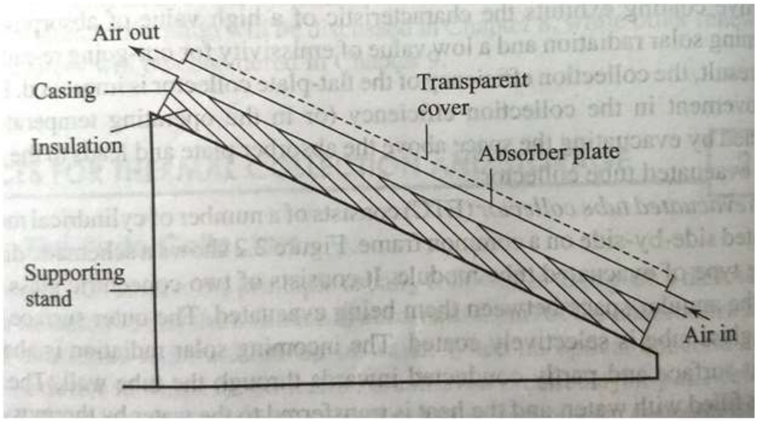
<td>Figure given below shows the schematic arrangement of a Solar Air Heater.



It consists of an absorber plate on which the solar radiation falls after coming through a transparent cover (usually made of glass). The absorbed radiation is partly transferred to the air flowing through tubes which are fixed to the absorber plate or are integral with it. This energy transfer is the useful gain. The remaining part of the radiation absorbed in the absorber plate is lost by convection and re-radiation to the surroundings from the top surface, and by conduction through the back and the edges. The transparent cover helps in reducing the losses by convection and re-radiation, while thermal insulation on the back and the edges helps in reducing the conduction heat loss.</td>
<td>4

3</td>
<td></td>
<td>7</td>
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<tr>
<td>VII</td>
<td>Show the working of a Line focusing parabolic collector with the help of a neat diagram.</td>
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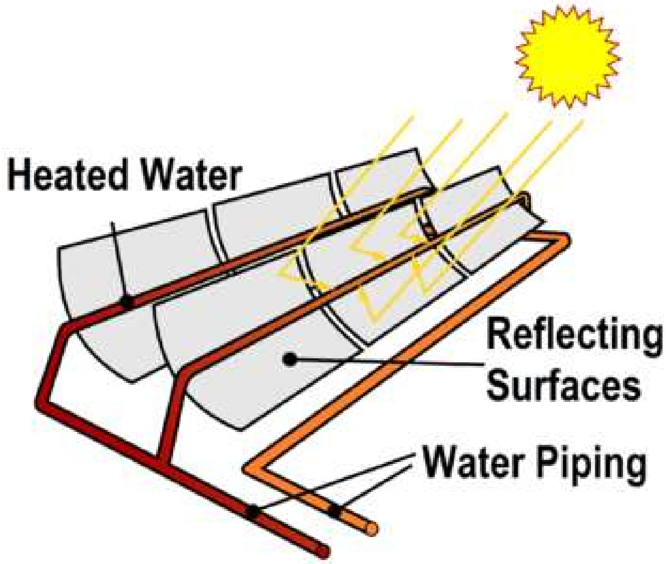
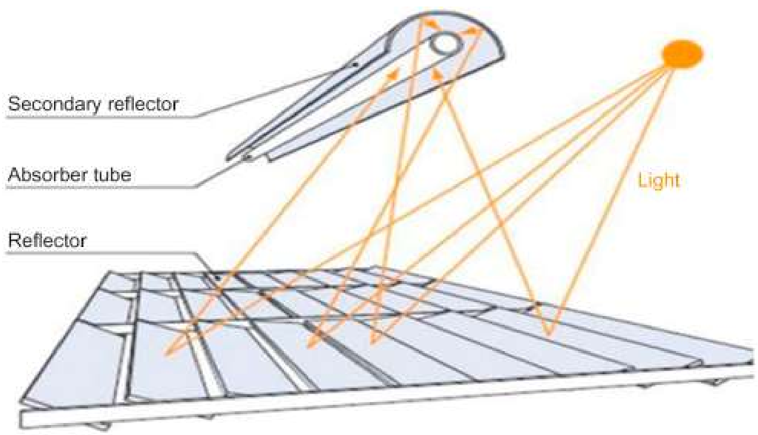
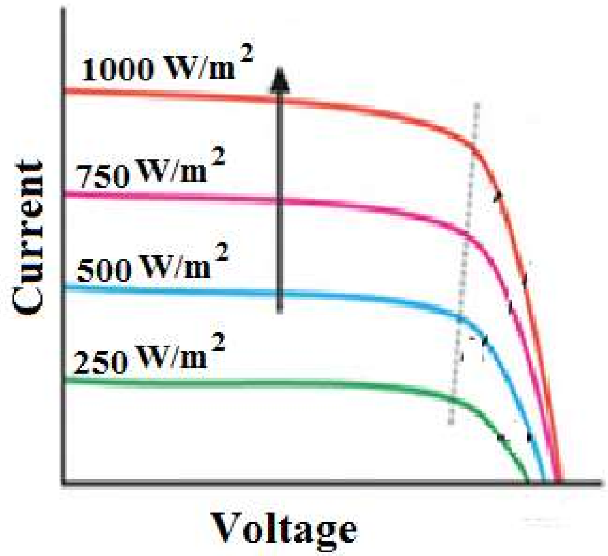
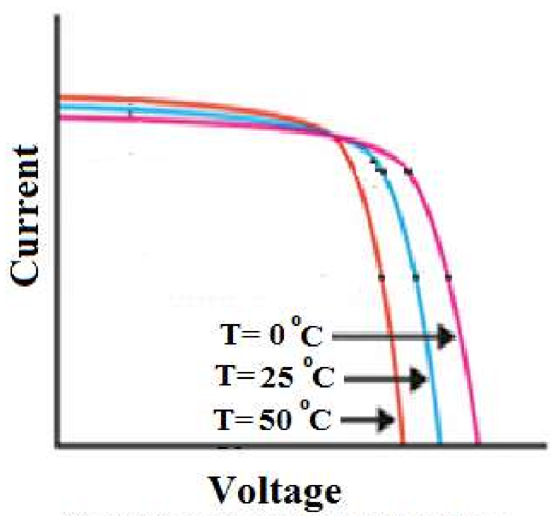
<td>7</td>
</tr>
</table>

	 These collectors use highly reflective materials to collect and concentrate the heat energy from solar radiation. These collectors are composed of parabolically shaped reflective sections connected into a long trough. A pipe that carries a fluid (like mineral oil, synthetic oil or molten salt) is placed in the center of this trough so that sunlight collected by the reflective material is focused onto the pipe, heating the contents to around 400 degree celsius. The heat is transferred to water in heat exchangers to generate steam for Solar thermal power plants.	4 3		7
	OR			
VIII	Show the construction of Fresnel reflectors with the help of a neat diagram.			

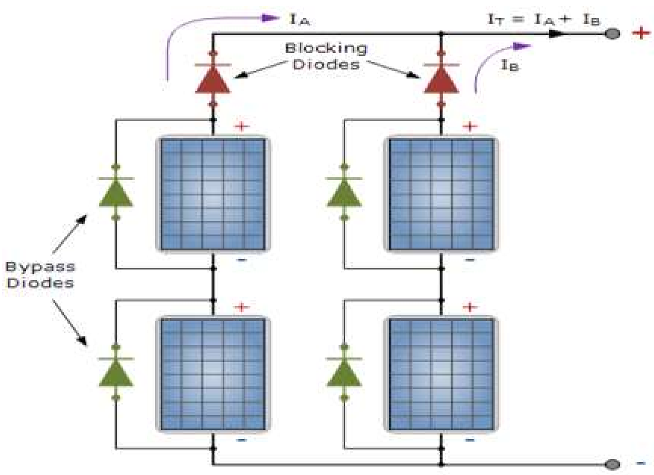
	A Linear Fresnel Reflector is a type of solar power collector. It uses flat mirrors as opposed to parabolic mirrors that are used in solar parabolic troughs. The basic principle remains the same with the mirrors collecting solar power which is then utilized to generate steam which in turn drives a turbine. This technology leads to the production of steam directly and does not use heat transfer fluid or other medium. The sunlight that is concentrated with the help of mirrors boils the water which is present in the receiver tubes thereby generating steam. No heat exchangers are used in this system. 	3 4		7
IX	A solar photovoltaic module having an area of 1.5 meter square gives a current and voltage of 8A and 28 V respectively, at maximum power point. The short circuit current of the module is 10 A and open circuit voltage is 32V. Find fill factor, maximum power point and efficiency of the solar cell.			7

	Short circuit current I_{sc} = 10 A Open circuit voltage, V_{oc} = 32 V Current at maximum power point, I_m = 8 A Voltage at maximum power point, V_m = 28 V Area = 1.5 m^2 Assuming STC, radiation intensity = 1000W/m^2 Hence input power = 1000 x 1.5 = 1500 W Maximum power point, $P_m = V_mI_m$ = 28 x 8 = 224 W Fill factor = $P_m/(V_{oc}$ x $I_{sc})$ = 224/(32 x10) = 0.7 Efficiency = P_m/Input power = 224/1500=0.149=14.9%	1 2 2 2		7
	OR			
X	Illustrate the variations in output voltage and current of a PV module with the changes in a) Irradiance b) Cell temperature			7

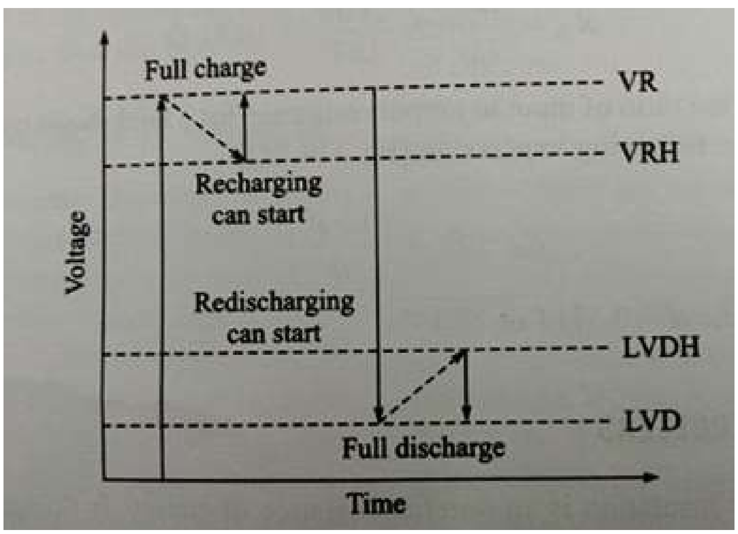
<table>
<tr><td></td><td>

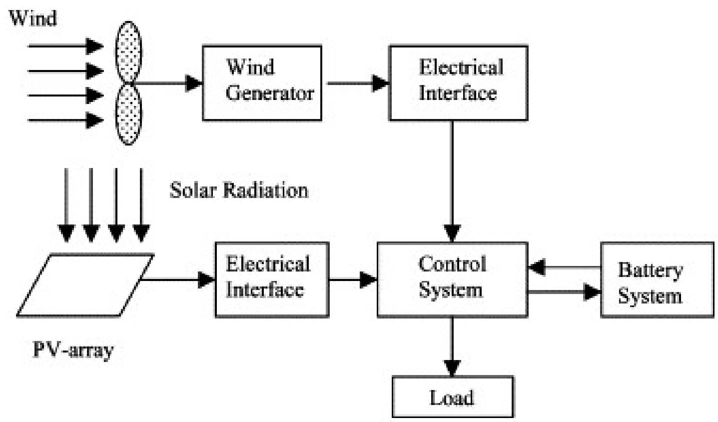
Variation of current and voltage with irradiance at constant temperature</td><td>2.5</td><td></td><td>7</td></tr>
<tr><td></td><td>From the figure it is clear that as irradiance increases current of a PV module increases, but the output voltage remains more or less constant.</td><td>1</td><td></td><td></td></tr>
<tr><td></td><td>

Variation of current and voltage with temperature of PV Module</td><td>2.5</td><td></td><td></td></tr>
<tr><td></td><td>From the above figure it is clear that current is directly proportional to temperature, whereas voltage is inversely proportional to temperature. Variation of voltage is effectively more than that of current and thereby power output decreases with increase in temperature.</td><td>1</td><td></td><td></td></tr>
</table>

XI	Explain the purpose of bypass and blocking diodes in solar PV systems.			7
	 **Blocking Diodes** The PN-junction diode acts like a solid state one way electrical valve that only allows electrical current to flow through themselves in one direction only. In the night when modules are not producing power, they can become loaded to batteries, i.e. current will flow in reverse direction .Blocking diodes can be used to block the flow of electric current from other parts of an electrical solar circuit. When used with a photovoltaic solar panel, these types of silicon diodes are generally referred to as Blocking Diodes. **Bypass Diodes** are used in parallel with either a single or a number of photovoltaic solar cells to prevent the current(s) flowing from good, well-exposed to sunlight solar cells from overheating and burning out weaker or partially shaded solar cells by providing a current path around the bad cell. Bypass diodes in solar panels are connected in “parallel” with a photovoltaic cell or panel to shunt the current around it, whereas blocking diodes are connected in “series” with the PV panels to prevent current flowing back into them. Blocking diodes are therefore different from bypass diodes although in most cases the diode is physically the same, but they are installed differently and serve a different purpose.	3 2 2		7

	OR			
XII	Explain the purpose of charge controllers and commonly used set points?			7

				7
	Purpose of charge controllers and commonly used set points	1		
	When PV systems are used for stand alone applications ,a backup power is necessary to compensate for the balance power demand of the load. Battery feeds the load when the PV output power is more than the load demand. When batteries are used, it is critical to prevent over charging or deep charging of the batteries to preserve their life and good performance. This is achieved by using charge controllers. A charge controller senses the voltage of the battery and decides either to disconnect it from the source to prevent it from overcharging or to disconnect the load to prevent deep discharging.			
	 	2		
	The charge control algorithms has set points (threshold values)			
	Regulation set point (VR): This set point is the maximum voltage a controller allows the battery to reach. At this point a controller will either discontinue battery charging or begin to regulate the amount of current delivered to the battery.	1		
	Regulation hysteresis (VRH): The set point is voltage span or difference between the VR set point and the voltage when the full array current is reapplied. The greater this voltage span, the longer the array current is interrupted from charging the battery. If the VRH is too small, then the control element will oscillate, inducing noise and possibly harming the	1		

XIII	Explain the working of a Solar-wind hybrid system with the help of a block diagram.			7
	 Hybrid systems are basically an integration of solar panels and wind turbines, the output of this combination is used to charge batteries, this stored energy can then be transmitted to local power stations. In this system wind turbines can be used to produce electricity when wind is available and solar panels are used when solar radiations are available. Power can be generated by both the sections at the same time also.	4 3		7
	OR			
XIV	Classify solar PV systems based on grid connectivity and storage.			7

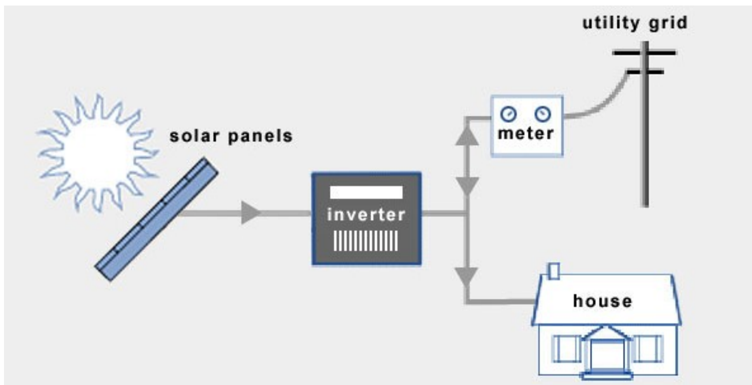
1. **Grid-tied system** 7

2. **Grid-tied system with battery back-up**

3. **Off-grid system**

1

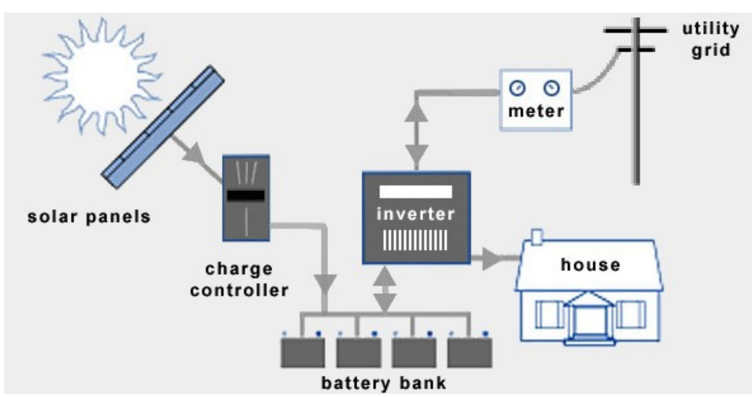
Grid-tied system



1

A grid-tied system is a basic solar installation that uses a standard grid-tied inverter and does not have any battery storage. Grid-tied systems are simple to design and are very cost effective because they have relatively few components. The main objective of a grid-tied system is to 1
lower your energy bill and benefit from solar incentives.

Grid-tied system with battery back-up



1

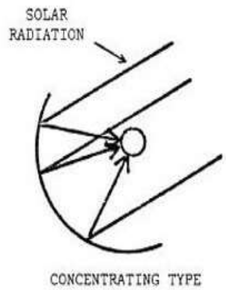
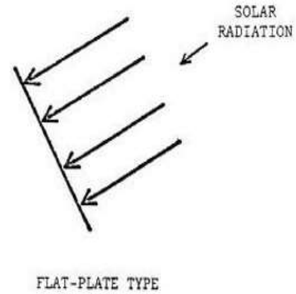
Grid tied system with battery back-up, otherwise known as a grid-hybrid system. This type of system is ideal for customers who are already on the grid who know that they want to have battery back-up. Battery based grid-tied systems provide power during an outage and you can store

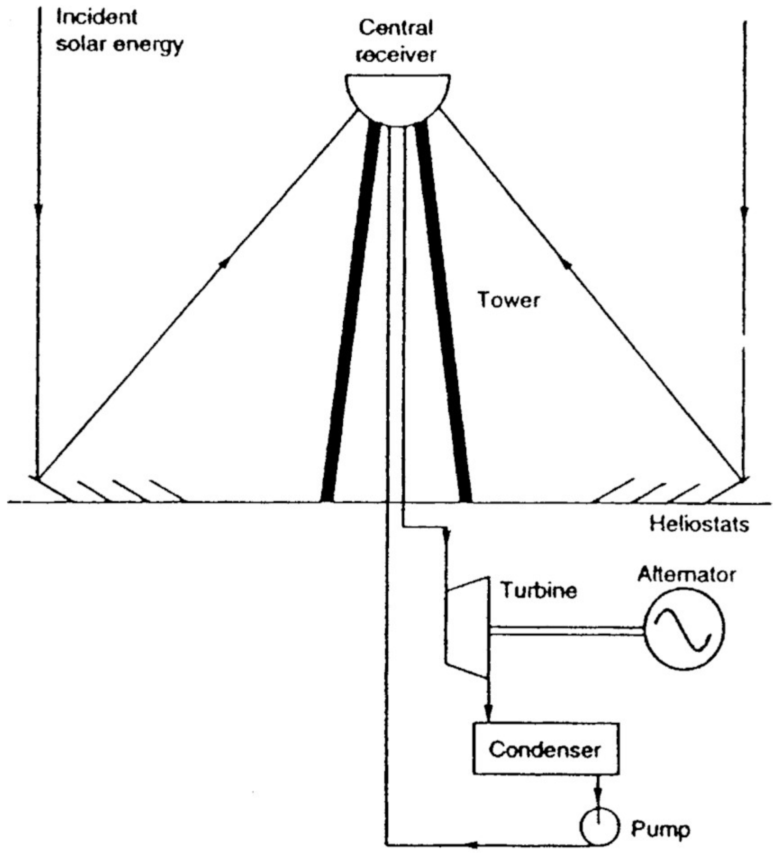
Scoring Indicators

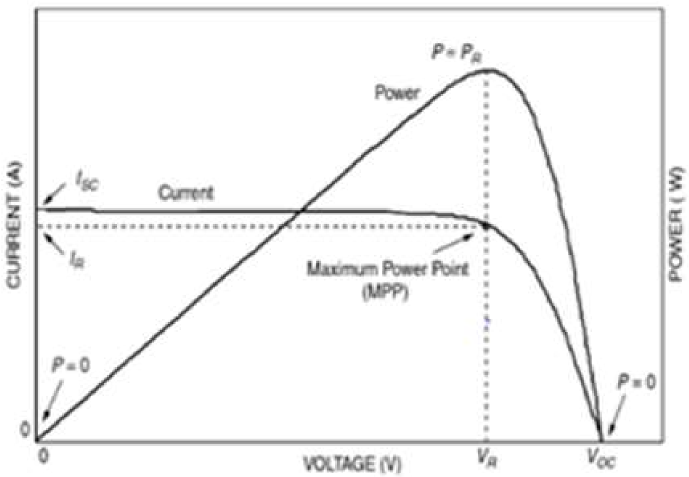
Model Question Paper II

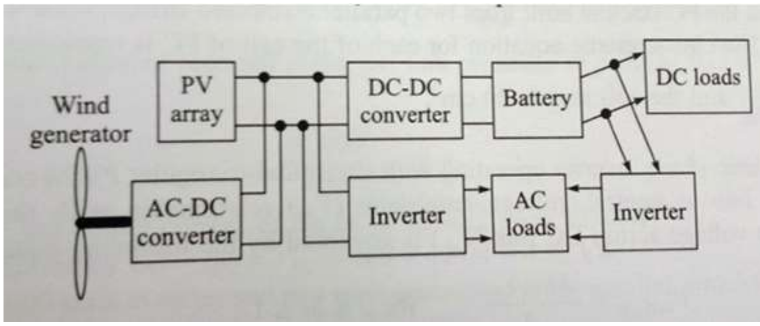
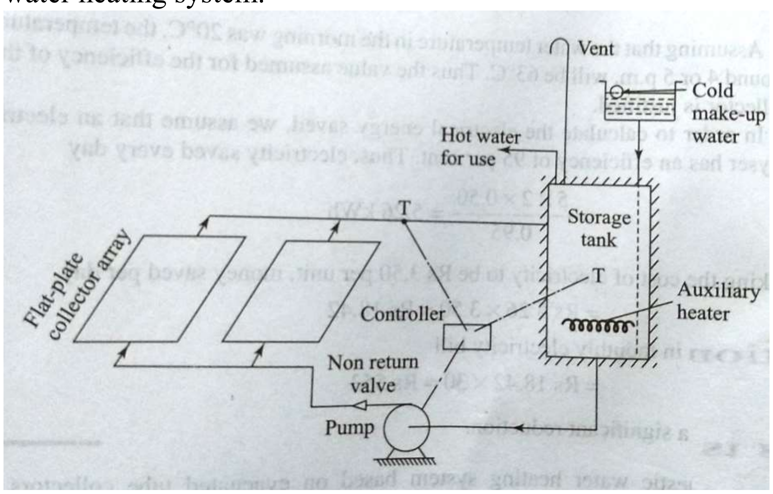
SOLAR POWER TECHNOLOGIES

Q No	Scoring Indicators	Split score	Sub Total	Total score
	PART A			**1**
I. 1	Concentrating collectors			1
I. 2	Dish type solar cooker			1
I. 3	Flat plate collector			1
I. 4	Parabolic dish or Fresnel reflectors			1
I.5	Photovoltaic effect			1
I. 6	The ratio of maximum power from a solar cell to the product of Voc and Isc.			1
I. 7	NiCad(Nickel-Cadmium)			1
I. 8	THD(Total Harmonic Distortion) is a measure of the amount of harmonics present in the output of a solar PV system.			1
I. 9	Boost converter is a dc to dc converter which steps up the input voltage to a higher voltage.			1
	PART B			
II. 1	Explain the potential of Solar energy in meeting our present and future energy demands in three points.			3
	1. Solar energy is a very large inexhaustible source of energy. 2. Unlike fossil fuels and nuclear power, it is an environmentally clean source of energy . 3. It is available in adequate quantities in almost all habitable parts of the world.	1 1 1		3
	Compare concentrating and non-concentrating type solar collectors.			3

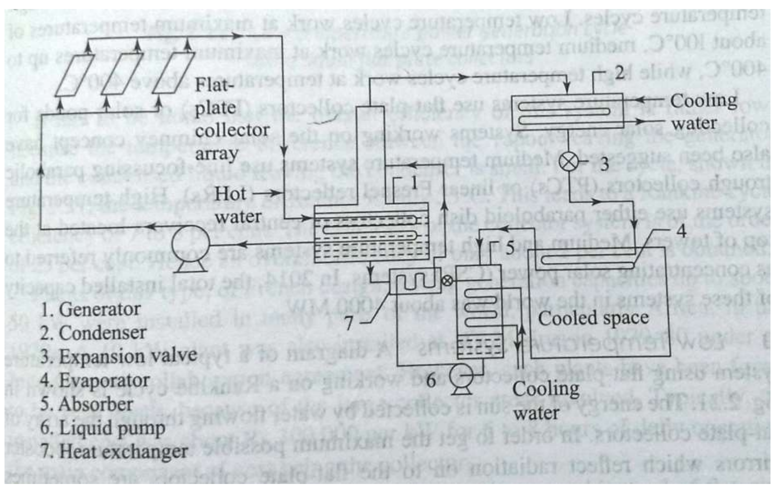
II. 2	<u>Concentrating type Solar collectors</u> 1. Concentrates the solar radiation over a large area to a small region. 2. High temperatures are obtained. 3.  <u>Non - concentrating type Solar collectors</u> 1. No concentration of radiation is done. 2. Low temperatures of the range of 40 to 100 degree celsius is obtained. 3.  	1 1 1		3
II.3	List any three characteristics of Passive Solar heating.			
	1. Passive solar heating does not use pumps, blowers or other mechanical devices. 2. It requires a special building design. 3. It is less expensive than an active system to construct and operate.	1 1 1		3
II.4	Classify concentrated solar power plants on the basis of temperature.			3
	1. Low temperature concentrated solar power plant (40 to around 100 degree celsius). 2. Medium temperature concentrated solar power plant (above 100 and up to 400 degree celsius). 3. High temperature concentrated solar power plant (Above 400 degree celsius).	1 1 1		3
II. 5	Draw the schematic diagram of a Central receiver solar power plant			3

		3		3
II.6	How is a solar PV module different from a solar PV cell? How can a solar PV module be made using individual solar cells?			
	A solar cell is the basic building unit of a solar PV module. A solar PV module is formed of an interconnection of solar cells in series and parallel. The ways in which interconnection of solar cells is obtained in a thin film technology and in a wafer-based technology are different. In thin film technology, cells are interconnected during the process of manufacturing of solar cells. While in wafer based technology, solar cells are manufactured first and then are interconnected to make PV modules.	1 2		3
	Draw the PV curve and VI curve of a photovoltaic module			3

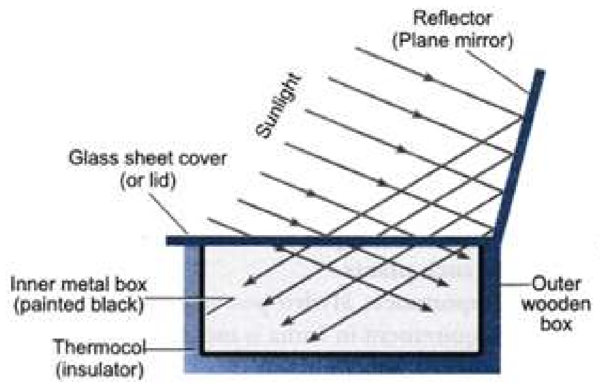
II. 7		3		3
II. 8	A Solar PV module has rated power of 100 W at STC. Its V_{oc} = 22V and I_{sc} = 5.75 A. Find out the fill factor. If the radiation intensity changes to 400 W/m2, will there be a change in the rated power output of the PV module? If yes, mention whether the rated power increases or decreases.			3
	It is given that rated power = 100 W, V_{oc} = 22V and I_{sc} = 5.75A Hence, $V_m X I_m$ = 100 W Fill factor = $(V_m X I_m)/(V_{oc} X I_{sc})$ = 100/(22 X 5.75)=0.79 Yes, there will be a change in rated power of the PV module. Rated power will decrease.	1 1 1		3
II. 9	Explain the main features of Net metering.			3
	Net metering is a billing mechanism that credits solar energy customers for the electricity they add to the grid. For instance, if you are using Solar energy to produce electricity at your home, it may generate more power than what your home needs during day time. If your home has the advantage of net metering, it allows you to send your extra power to the grid and provide credit that you can use when you need it. Net metering ensures the energy you generate at home doesn't go in waste.	3		3
II.10	Draw a neat block diagram of a solar wind hybrid system.			3

	 	3		3
	PART C			
III	Explain the operation of a forced circulation solar water heating system with the help of a neat schematic diagram.			7
	Figure shows the schematic diagram of a forced circulation water heating system.  Water from the storage tank is pumped through the collector array, where it is heated and then made to flow back into the storage tank. Whenever hot water is withdrawn for use, cold make-up water takes its place. The pump for maintaining the forced circulation is operated by an on-off controller which senses the difference between the temperature of the water at the exit of the collectors and a suitable location inside the storage tank. The pump is switched on whenever this difference exceeds a certain value and the pump is switched off when it falls below a certain value.	3 4		7

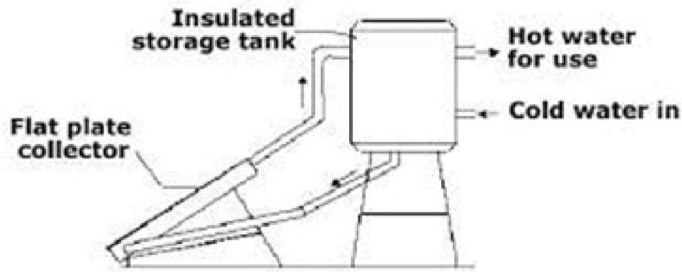
<table>
<tr><td></td><td>OR</td><td></td><td></td><td></td></tr>
<tr><td>IV</td><td>Explain Solar space cooling with the help of a neat schematic diagram.</td><td></td><td></td><td>7</td></tr>
<tr><td></td><td>



Water heated in a flat plate collector array is passed through a heat exchanger called the generator, where it transfers heat to a solution mixture of the absorbent and refrigerant, which is rich in refrigerant. Refrigerant vapour is boiled off at a high pressure and goes to the condenser where it is condensed into a high pressure liquid. The high pressure liquid is throttled to a low pressure and temperature in an expansion valve, and passes through the evaporator coil. Here, the refrigerant vapour absorbs heat and cooling is therefore obtained in the space surrounding this coil. The refrigerant vapour is now absorbed into a solution mixture withdrawn from the generator, which is weak in refrigerant concentration. This yields a rich solution which is pumped back to the generator, thereby completing the cycle. The rich solution flowing from the absorber to the generator is usually heated in a heat exchanger by the weak solution withdrawn from the generator. This helps to improve the performance of the cycle.</td><td>3
4</td><td></td><td>7</td></tr>
<tr><td></td><td>Illustrate the operation of a Solar cooker.</td><td></td><td></td><td>7</td></tr>
</table>

V	A solar cooker is a device in which heat energy from sunlight is trapped inside the cooker using the phenomenon of greenhouse effect. It consists of a mirror, black box and glass lid. The mirror is mounted on the box to concentrate light rays onto the cooker box. The box is painted black from the inside to absorb the maximum light. It is covered with a glass lid to produce a greenhouse effect inside.  	3 4		7
VI	Explain the working of a solar water heater using natural circulation system.			7

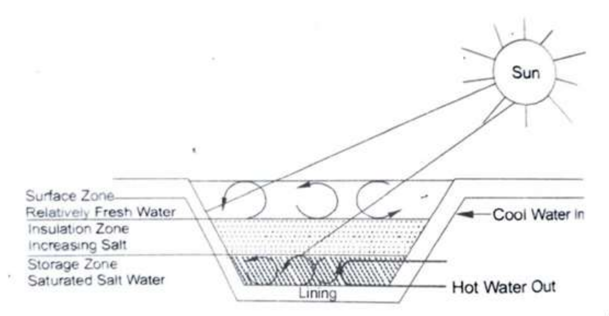
<table>
<tr><td></td><td>technique.



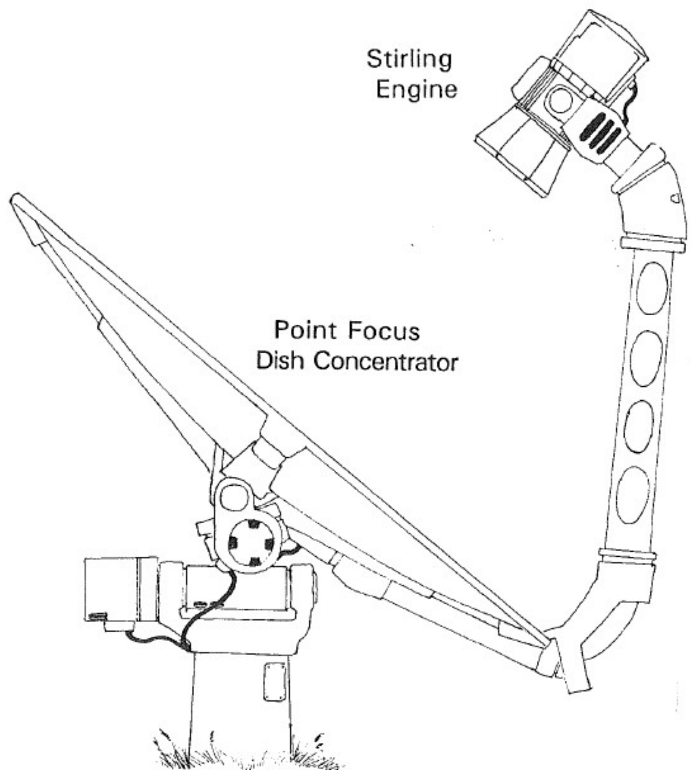
Small capacity natural circulation water heating system

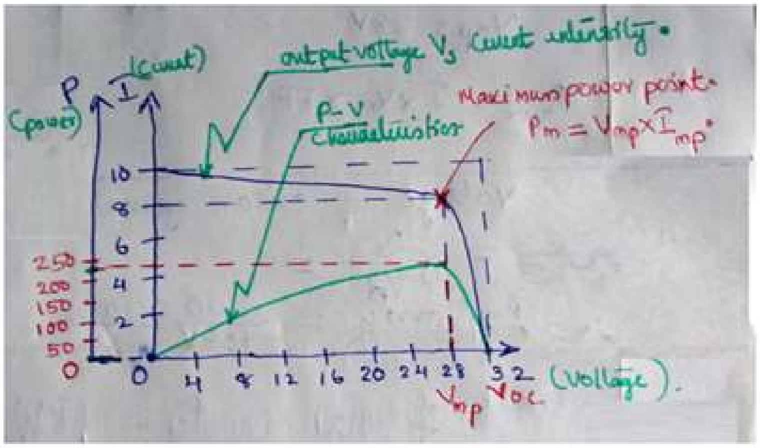
The two main components of the system are the liquid flat – plate collector and the storage tank, the tank being located above the level of the collector. As water in the collector is heated by solar energy, it flows automatically to the top of the water tank and it is replaced by cold water from the bottom of the tank. Hot water for use is withdrawn from the top of the tank. Whenever this is done, cold water automatically enters the bottom. An auxiliary-heating system is sometimes provided for use on cloudy or rainy days.</td><td>4

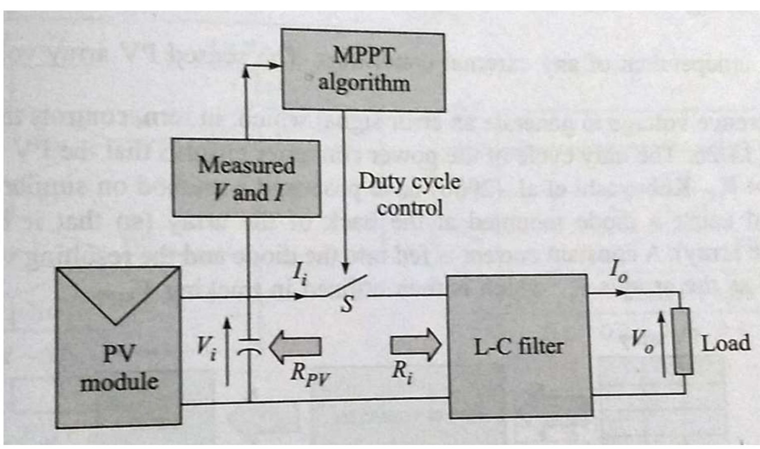
3</td><td></td><td>7</td></tr>
<tr><td>VII</td><td>Explain the working of a Solar Pond.</td><td></td><td></td><td>7</td></tr>
<tr><td></td><td>Solar pond is a system of collecting and storing solar energy. A solar pond is an artificially constructed pond in which significant temperature rises are caused to occur in the lower regions by preventing convection. The pond water is hot enough for space heating and other applications. In the salt gradient solar ponds, dissolved salt is used to create layers of water with different densities.


</td><td>3

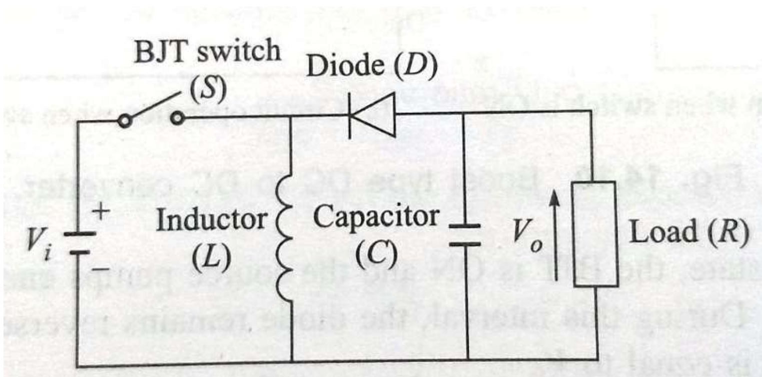
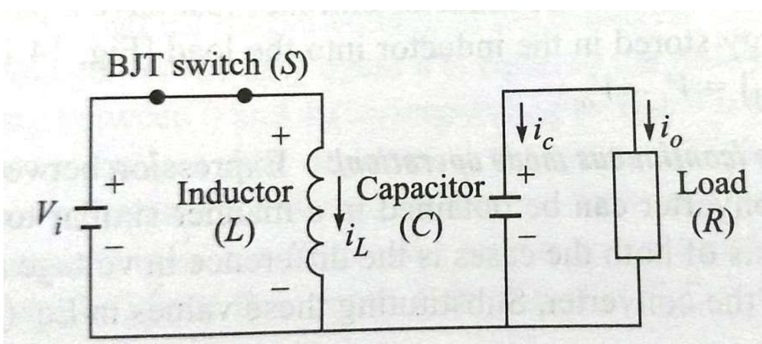
4</td><td></td><td>7</td></tr>
<tr><td>VIII</td><td>Explain the working of a Solar Powered Stirling Engine.</td><td></td><td></td><td></td></tr>
</table>

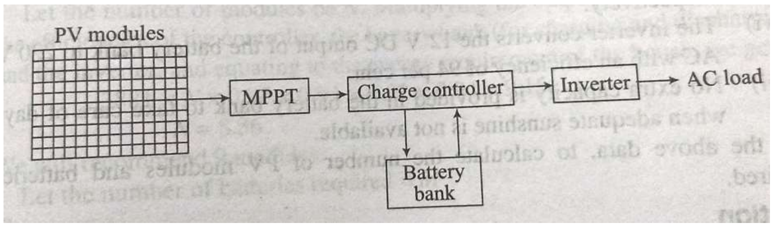
	Stirling Engine is an External Combustion Engine which converts thermal energy into mechanical energy. This mechanical energy can be further converted to electrical energy. The thermal energy required in a Stirling engine can be produced using Solar energy. The apparatus consists of a parabolic dish which concentrates the solar energy received to a focal point at the center of the dish. The concentrated solar energy is used to drive the Stirling Engine.  	3 4		7
IX	The Fill factor of a PV panel is 0.7. Draw the PV characteristics and mark Vmp if Voc= 32V, Isc= 10A, Imp = 8A			7

	Fill factor=(Vmp Imp)/(Voc Isc)=0.7 Isc=8A, Voc=32V and Imp=8A 0.7= (Vmp*8)/(32*10) Vmp=28V 	3 4		7
	OR			
X	A solar photovoltaic module having an area of 1.62 meter square gives a current and voltage of 7.83A and 29.4V respectively, at maximum power point. The short circuit current of the module is 8.52A and open circuit voltage is 36.7V. Find fill factor, maximum power point and efficiency of the solar cell.			7
	Short circuit current I_{sc} = 8.52 A Open circuit voltage, V_{oc} = 36.7 V Current at maximum power point, I_m = 7.83 A Voltage at maximum power point, V_m = 29.4 V Area = 1.62 m^2 Assuming STC, radiation intensity = 1000W/m^2 Hence input power = 1000 x 1.62 = 1620 W Maximum power point, $P_m = V_m I_m$ = 29.4 x 7.83 = 230.202 W Fill factor = $P_m/(V_{oc}$ x $I_{sc})$ = 230.202/(8.52 x 36.7) = 230.202/312.712 = 0.736 Efficiency = P_m/Input power = 230.202/1620=0.142=14.2%	1 2 2 2		7

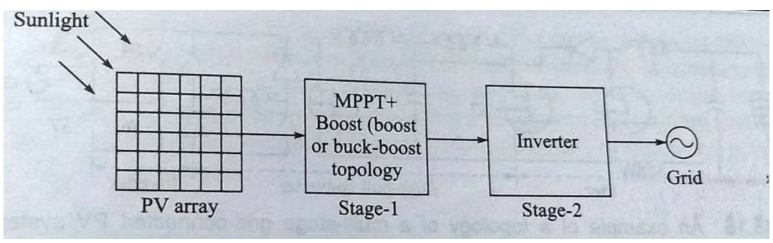
XI	Explain MPPT with the help of a block diagram.			7
	Maximum Power Point Tracking (MPPT) is a method employed in Solar PV systems to ensure that maximum power transfer takes place between PV modules to the load. When a solar PV module is used in a system, its operating point is decided by the load to which it is connected. Also, since solar radiation falling on a PV module varies throughout the day, the operating point of the module also changes throughout the day. Ideally, under all operating conditions, it is desired to transfer maximum power from the PV module to the load. In MPPT, electronic circuitry is used to ensure that the maximum amount of generated power is transferred to the load. Maximum power tracking mechanism makes use of an algorithm and an electronic circuitry. The mechanism is based on the principle of impedance matching between load and PV module. Impedance matching is done by using a DC to DC converter in which the duty cycle of the switch is changed. Consider the block diagram ofMPPT algorithm shown below.  Power from the solar module is calculated by measuring the voltage and current. This power is an input to the algorithm which then adjusts the duty cycle of the switch resulting in the adjustment of the reflected load impedance to match that of the impedance of the Solar PV module in a given operating condition for maximum power transfer.	1 1 1 2 2		
	OR			

XII	Explain the working of a Buck-boost type DC to DC converter used in a solar photovoltaic power plant with the help of circuit diagrams.			7

	Figure given below shows the circuit diagram of a Buck-boost converter.  It is a type of DC to DC converter that has an output voltage magnitude that is either greater than or equal to or less than the input voltage magnitude. The output voltage is adjustable based on the duty cycle of the switching transistor. It has two states of operation: (i) ON state and (ii) OFF state	3		7
	(i) ON state:  While in the ON state, the BJT is ON and conducting and the input voltage source is directly connected across the inductor as shown. This results in accumulation of energy in L. During this interval diode D is reverse biased and the capacitor supplies energy to the load. (ii) OFF state:	2		

<table>
<tr><td>XIII</td><td>Explain stand-alone Solar PV systems with the help of a neat block diagram.</td><td></td><td></td><td>7</td></tr>
<tr><td></td><td>

An MPPT unit operates the PV array at an optimum power level. The power is then stored in a battery bank through a controller which regulates both the charging and discharging process from the batteries. The DC output of the battery bank is passed through the inverter and converted to the AC voltage required by the load.</td><td>3

4</td><td></td><td>7</td></tr>
<tr><td></td><td>OR</td><td></td><td></td><td></td></tr>
<tr><td>XIV</td><td>Explain grid connected Solar PV systems with neat block diagrams. Also mention one advantage of grid connected Solar PV systems.</td><td></td><td></td><td>7</td></tr>
<tr><td></td><td>

The figure shows a grid-connected PV system with two stages to process PV power and feed into the grid. The two stages help in appropriately conditioning the available solar power for feeding into the grid. The first stage is used to boost the PV array voltage and track the maximum solar power, the second stage inverts this DC power into high quality AC power.
An advantage of grid connected PV systems is that they do not need batteries. This helps in reducing the cost.</td><td>3

3

1</td><td></td><td>7</td></tr>
</table>